A User Experience Study on Verification of their existence for pensioners

A System That Doesn’t Speak Back?

This project explores how clearer feedback and guided interactions can improve the fingerprint biometric verification experience.

“This Project Started With a conversation”

It started with my grandfather, who spent an entire day trying to complete a simple life certificate submission.

“I kept trying to scan my finger… it just didn’t tell me what I was doing wrong.”

That’s when it clicked-he wasn’t struggling with the task,
he was struggling with a system that said nothing back.

What is a life certificate?

A life certificate is a verification process where a person confirms they are alive, so their payments can continue without being stopped.

“I Thought It Was a Product Problem… Until the Data Said Otherwise”

Initial Assumption

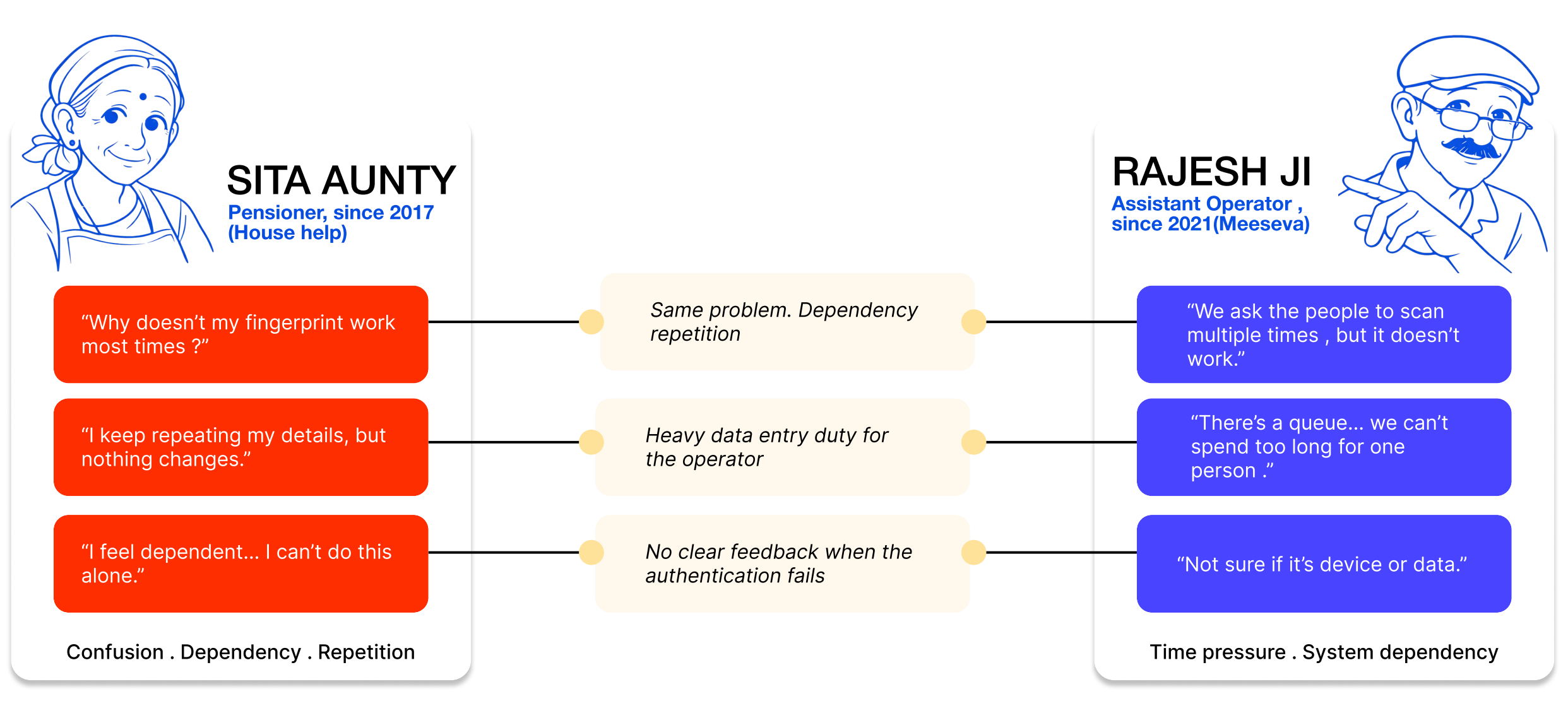
- The issue lies in the fingerprint scanner (product)
- Improve hardware accuracy
- Improve scan sensitivity

So I started investigating…

I realized it takes more then one person to make the experience happen, *not only the pensioner but also the operator who assists.*

so i tried to connect both the pensioner’s and operator’s pressure points.

From multiple interviews i viewed, 2 real conversations that proved my assumptions wrong



SITA AUNTY
Pensioner, since 2017 (House help)

RAJESH JI
Assistant Operator , since 2021(Meeseva)

SITA AUNTY		RAJESH JI
“Why doesn’t my fingerprint work most times ?”	*Same problem. Dependency repetition*	“We ask the people to scan multiple times , but it doesn’t work.”
“I keep repeating my details, but nothing changes.”	*Heavy data entry duty for the operator*	“There’s a queue… we can’t spend too long for one person .”
“I feel dependent… I can’t do this alone.”	*No clear feedback when the authentication fails*	“Not sure if it’s device or data.”
Confusion . Dependency . Repetition		Time pressure . System dependency

The problem wasn’t the product.
It was the interaction between user, system, and guidance.

Why This, Why Now

This didn’t start as a system problem. It started with a person

Watching my grandfather and many others like him struggle through something as essential as submission of life certificate to receive pension made me realize how much the experience was *failing the very people it was meant to support.*

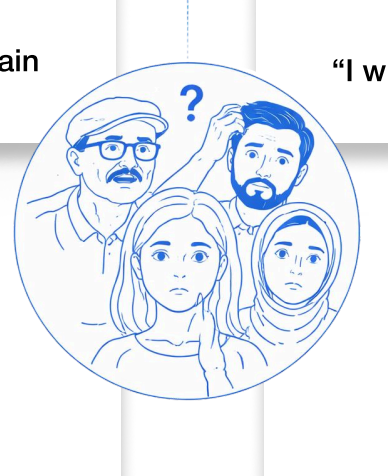
I decided to look deeper into this because I wanted to:

- “Make verification simple (not a guessing game)”
- “Reduce the ‘try again… and again’ loop”
- “10–15% of biometric transactions fail due to fingerprint mismatch
- “Reduce dependency on the CSC operators”

People weren’t just failing the system-the system wasn’t guiding them to succeed.

“What happens when the system refuses to talk back?”

Interview insights were mapped to understand users’ thoughts, emotions, and behaviors through empathy mapping



SAYS
- “Why is my fingerprint not working?”
- “I keep trying until it accepts.”
- “Can someone help me?”
- “The machine doesn’t explain anything.”

THINKS
- “Am I doing this correctly?”
- “What if it fails again?”
- “I hope this works.”
- “I wish it guided me.”

DOES
- Retries fingerprint
- Adjusts finger position
- Waits for system response
- Seeks help from others

FEELS
- Anxious
- Confused
- Frustrated
- Relieved when it works

In many cases, users didn’t just retry, they turned to operators for help, reinforcing dependency.

The Questions that Shaped Everything

What is the real problem here?
- Lack of clear feedback and high dependency on operators

Where do users struggle the most?
- Biometric verification and assisted data entry

Why does this matter now?
- Failures increase anxiety and force reliance on operators

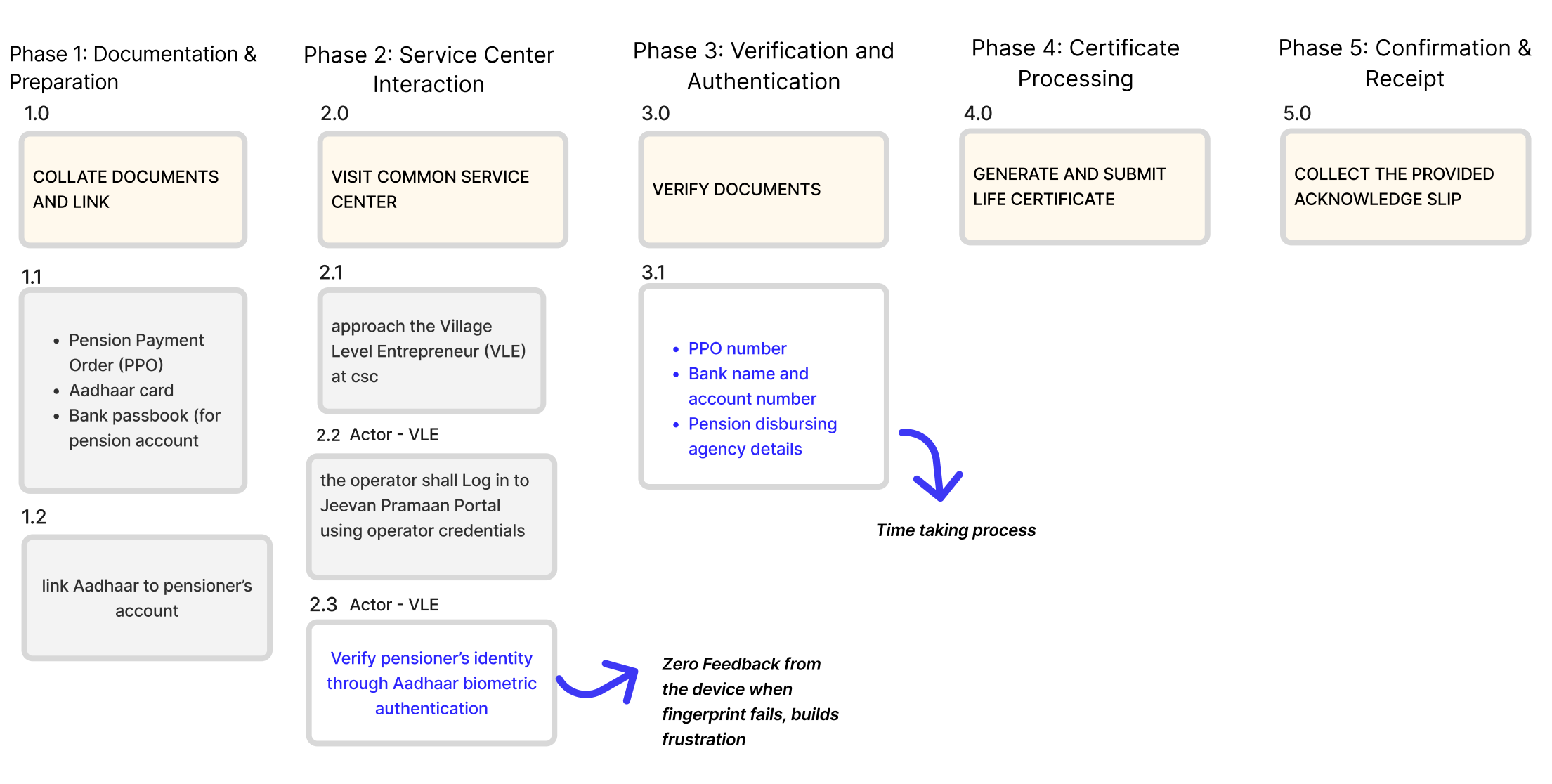
How can this be improved?
- By guiding users clearly and enabling more independent interactions

These weren't rhetorical. Every design decision i took had to trace back to at least one of these.

Current life certificate submission flow

USER GOAL

`SUBMISSION OF LIFE CERTIFICATE THROUGH COMMON SERVICE CENTER`



Phase 1: Documentation & Preparation
1.0 COLLATE DOCUMENTS AND LINK
1.1
- Pension Payment Order (PPO)
- Aadhaar card
- Bank passbook (for pension account

1.2 link Aadhaar to pensioner’s account

Phase 2: Service Center Interaction
2.0 VISIT COMMON SERVICE CENTER
2.1 approach the Village Level Entrepreneur (VLE) at csc
2.2 Actor - VLE
the operator shall Log in to Jeevan Pramaan Portal using operator credentials
2.3 Actor - VLE
Verify pensioner’s identity through Aadhaar biometric authentication

Zero Feedback from the device when fingerprint fails, builds frustration

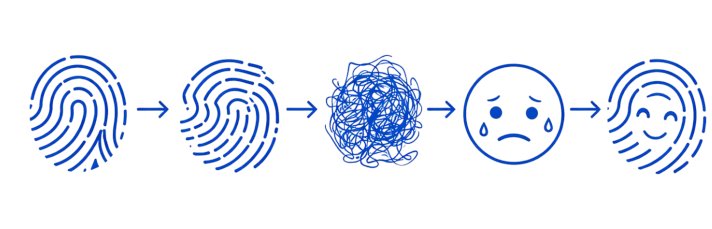
Phase 3: Verification and Authentication
3.0 VERIFY DOCUMENTS
3.1
- PPO number
- Bank name and account number
- Pension disbursing agency details

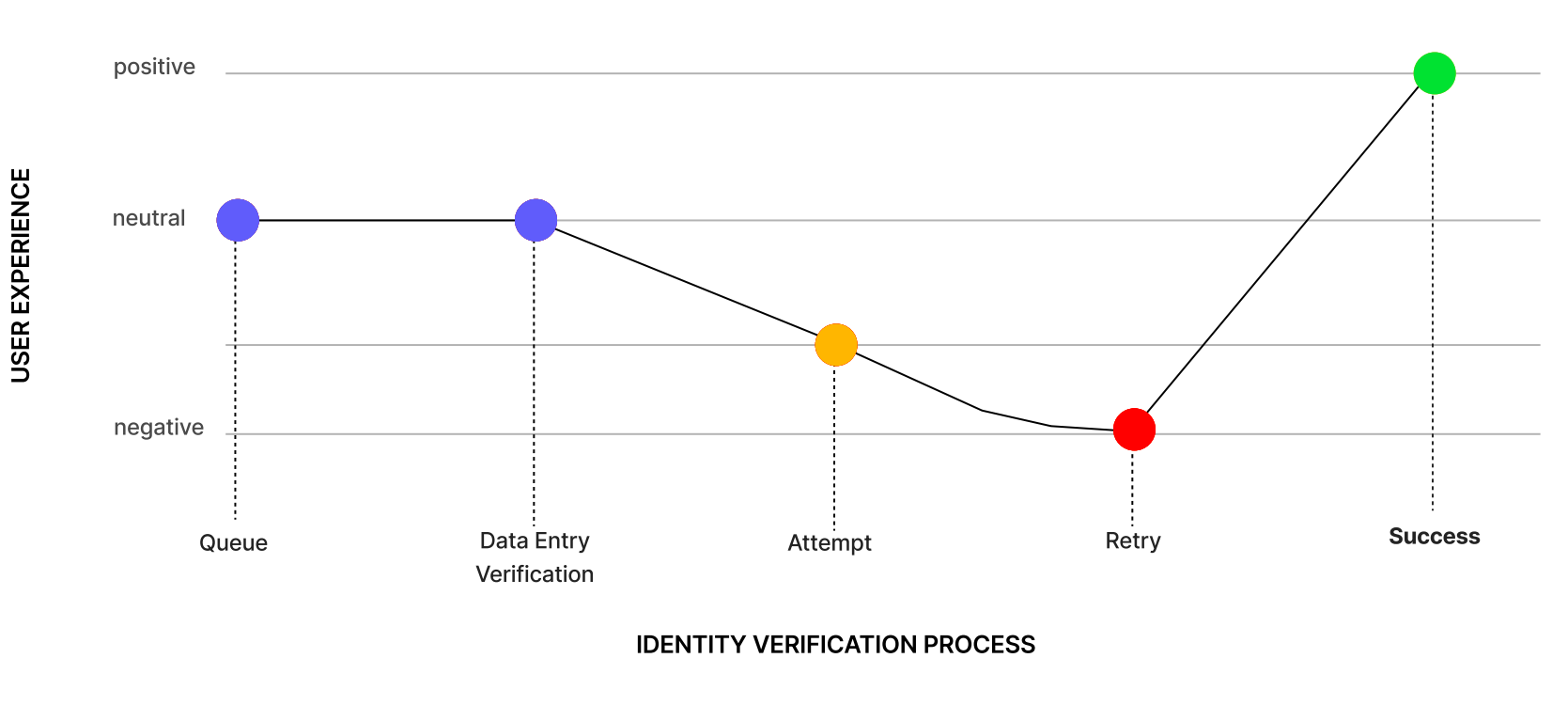
Time taking process

Phase 4: Certificate Processing
4.0 GENERATE AND SUBMIT LIFE CERTIFICATE

Phase 5: Confirmation & Receipt
5.0 COLLECT THE PROVIDED ACKNOWLEDGE SLIP

“Understanding User Experience When Feedback Breaks Down”





The journey starts off smooth but drops sharply during data entry and biometric verification, where repeated failures and lack of feedback create the most frustration.

“The problem wasn’t failure-it was not knowing why.”

Data	Majority of users require operator assistance to complete verification Multiple retries are common before success “10–15% of biometric transactions fail due to fingerprint mismatch
Finding	The process consistently broke down at biometric verification, where unclear feedback and repeated failures pushed users to depend on operators.
Insights	Users weren’t failing at the task-the system wasn’t giving them enough information to succeed independently, which led to repeated attempts and reliance on operators.

How might we enable pensioners to complete biometric verification independently with confidence?

Introducing: Meeseva Assisted Kiosk (at CSC Centres)
A guided, self-service biometric verification system designed for pensioners(Currently Targeting the Pain Points).

Problem-1 (Current System)

Manual Data Entry → Operator Assistance → Biometric Verification → Submission

Proposed System (Solution)

1. minimizing manual data entry

Biometric-First Auto Fetch

Biometric First → Auto-fetch Data → User Confirms

By prioritizing biometric authentication first, user data is automatically fetched, eliminating the need for manual entry and reducing reliance on operator assistance.

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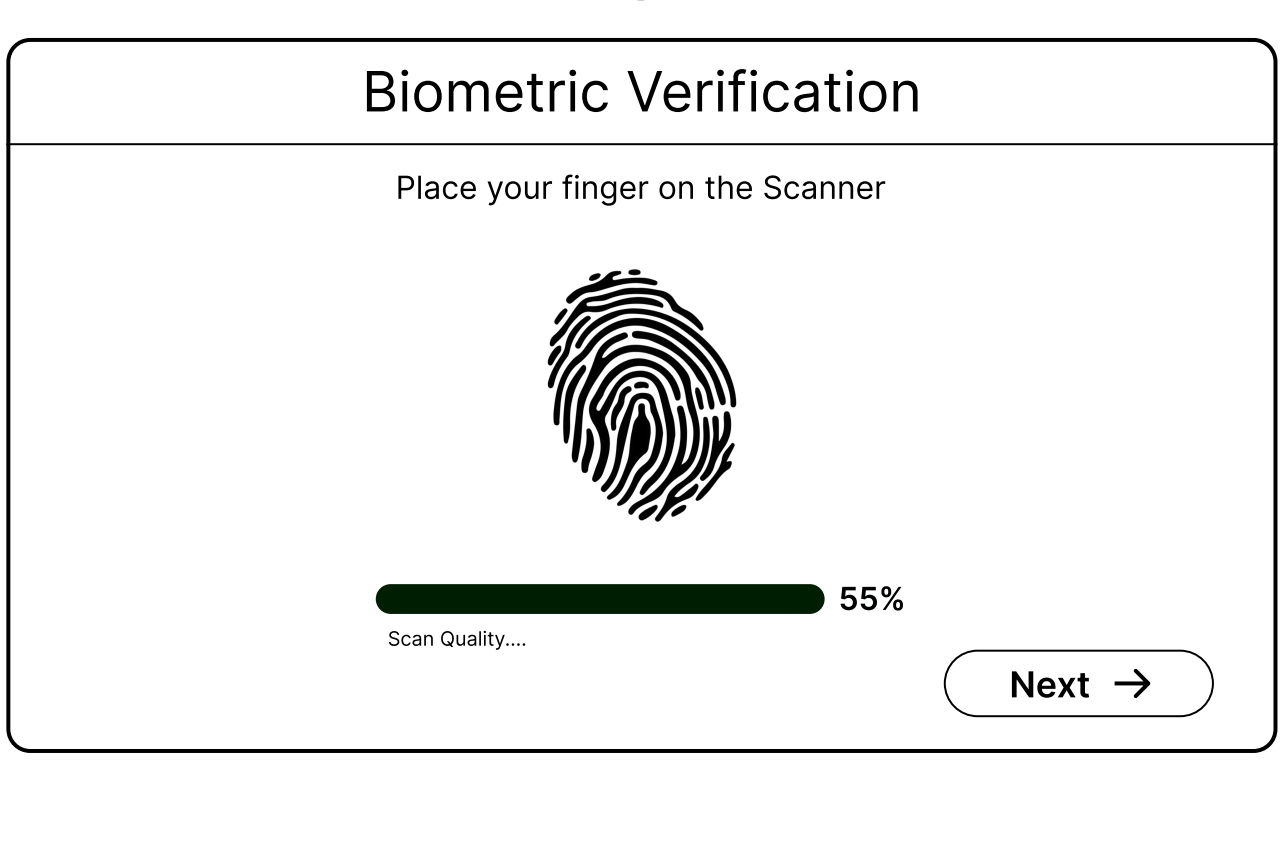
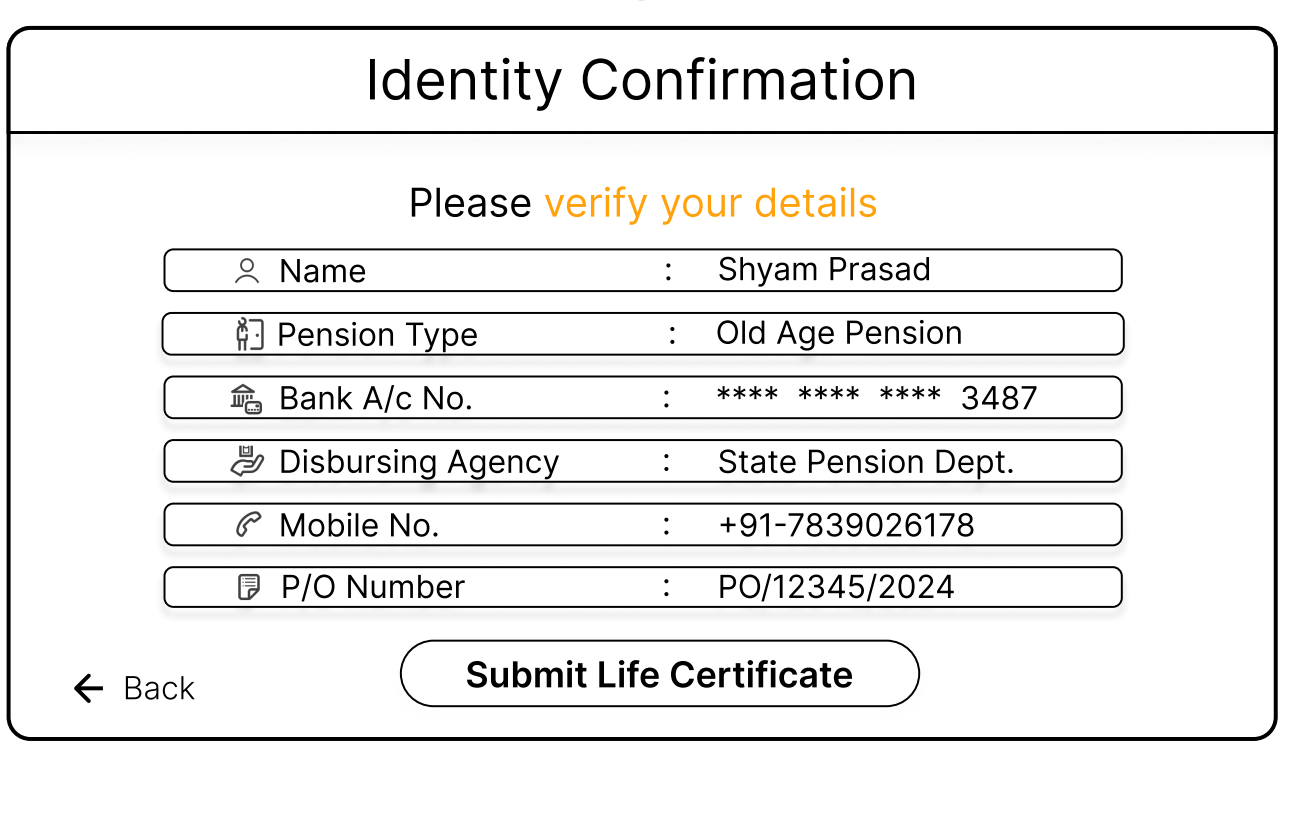


Biometric Verification
Place your finger on the Scanner
55%
Scan Quality...
Next →

②



Identity Confirmation

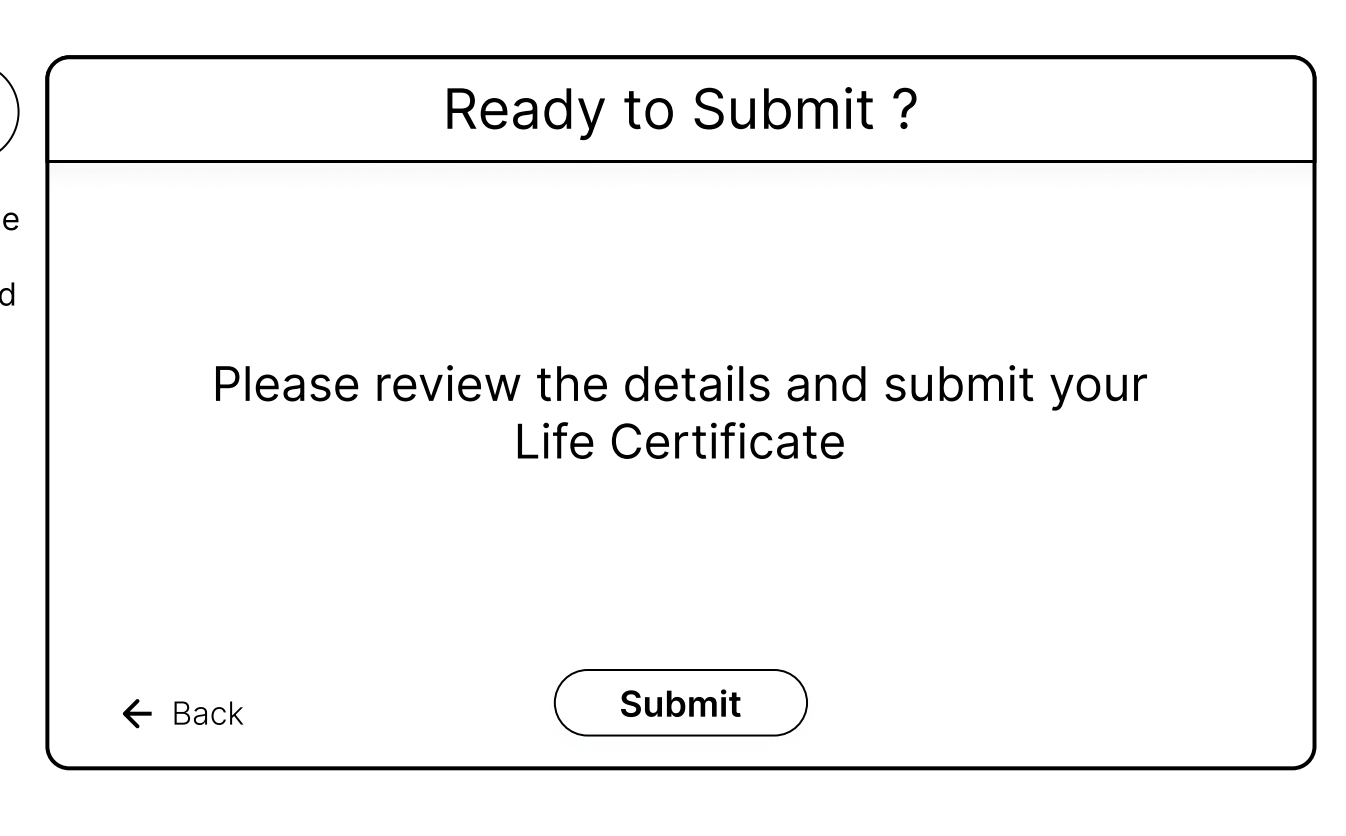
Please verify your details

Name	:	Shyam Prasad
Pension Type	:	Old Age Pension
Bank A/c No.	:	**** **** **** 3487
Disbursing Agency	:	State Pension Dept.
Mobile No.	:	+91-7839026178
P/O Number	:	PO/12345/2024

← Back
Submit Life Certificate

③

double confirmation as the details of the pensioners automatically gets fetched



Ready to Submit ?

Please review the details and submit your Life Certificate

← Back
Submit

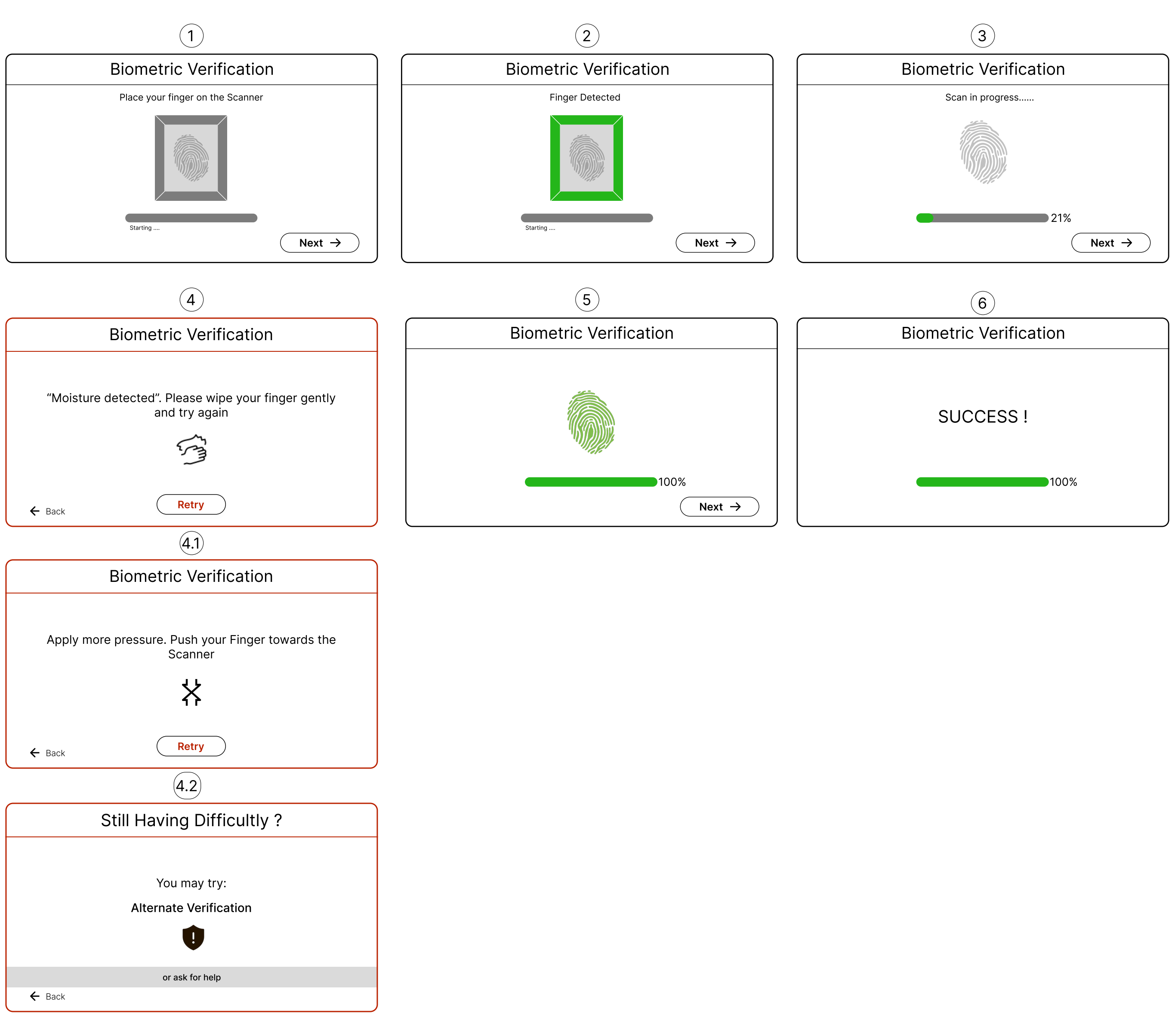
Problem-2 (Current System)

Place Finger → Scan Attempt → Failure/Unclear Result → Retry (No Guidance)

Proposed System (Solution)

Place Finger → Real-time Feedback → Guided Correction → Successful Scan

Your finger might be slightly moist. Try gently wiping it and place it again.
Press your finger a little more firmly on the scanner.
Try repositioning your finger so it covers the scanner properly.



① Biometric Verification
Place your finger on the Scanner
Starting ....
Next →

② Biometric Verification
Finger Detected
Starting ....
Next →

③ Biometric Verification
Scan in progress......
21%
Next →

④ Biometric Verification
“Moisture detected”. Please wipe your finger gently and try again
← Back
Retry

④.① Biometric Verification
Apply more pressure. Push your Finger towards the Scanner
← Back
Retry

④.② Still Having Difficultly ?
You may try:
Alternate Verification
or ask for help
← Back

⑤ Biometric Verification
100%
Next →

⑥ Biometric Verification
SUCCESS !
100%

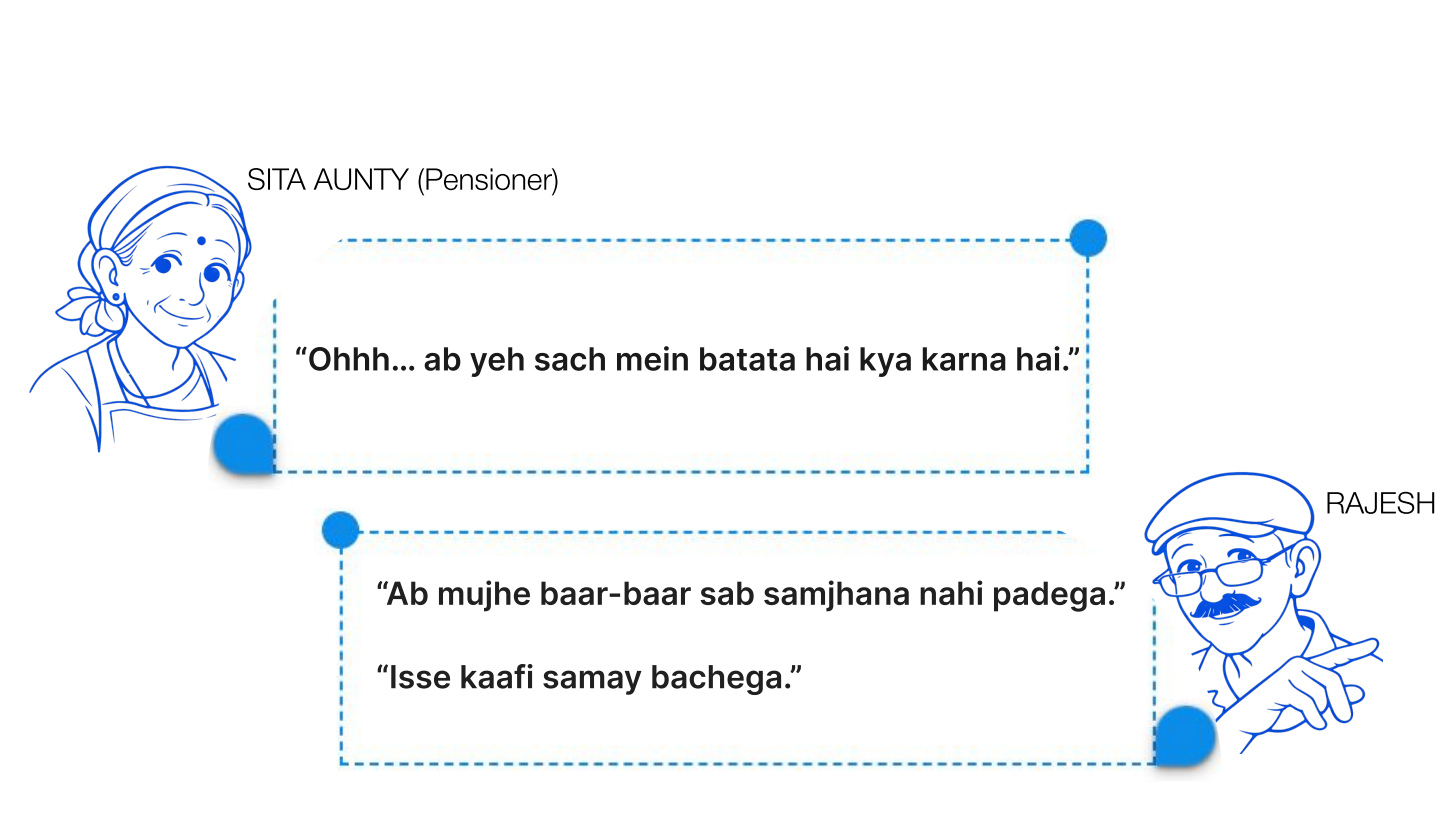
What started as a simple conversation became a deeper realization: the real barrier wasn’t technology, it was the lack of guidance around it.

Key Learnings

- *Failure is rarely the problem*-unclear feedback is.
- *Independence comes from guidance*, not simplification alone.
- Designing the experience has taught me to *look beyond the users*.

This project reframed biometric verification from a task to complete into an experience to support.

Soon after, i went back to explain the wireframe solutions to Sita aunty and Rajesh ji who i had previously interviewed .



SITA AUNTY (Pensioner)

“Ohhh... ab yeh sach mein batata hai kya karna hai.”

RAJESH JI (OPERATOR)

“Ab mujhe baar-baar sab samjhana nahi padega.”

“Isse kaafi samay bachega.”

This is just the starting point.

In Studio 2, I will take these insights forward into designing and testing solutions that enable truly independent, confident verification experience.

Thank you!

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